

DRAFT: Research: SASVA AI for QA Strategy



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***Purpose:** This page outlines a comprehensive study of how **SASVA (Persistent's Internal GenAI)** can be leveraged as a **Virtual QA Engineer** in automation and manual testing teams. It maps our internal QA workflows and test lifecycle tasks to SASVA capabilities and demonstrates how SASVA outperforms other AI tools like GitHub Copilot and ChatGPT for Persistent's needs.*

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Section 1: Tech Stack Alignment

The QA Automation team at GS currently uses:

Domain	Technology Tools
Language	Java (17+), Groovy
Frameworks	Selenium, Cucumber, TestNG
Build & CI/CD	Maven, Jenkins
Test Management	qTest, JIRA
Version Control	GitLab
Database Validation	Oracle SQL
DevOps Integration	Jenkins Pipelines
Cloud Integration	BrowserStack, Azure
Reporting	TestNG Reports, Console Logging
File/Logs	CSV, XML, Logs, Property Files

SASVA can integrate into all the above touchpoints using API connectors, automation pipelines, and conversational prompts.

Section 2: Where SASVA Can Help QA Teams

SASVA can become a **Virtual QA Engineer** across the STLC (Software Testing Life Cycle) as shown below:

1. Test Case Authoring & Optimization

- Generate or refactor **Given-When-Then** BDD-style Cucumber scenarios
- Auto-suggest **edge cases**, **negative test cases**, and **boundary validations**
- Refactor test cases based on updated feature descriptions in JIRA

2. Automation Script Assistance

- Convert manual test cases to **Java/Cucumber/TestNG** automation code
- Highlight reusable functions
- Recommend locator strategies using HTML/DOM context

3. Regression Suite Maintenance

- Auto-identify flaky tests based on execution logs
- Suggest **redundant or outdated tests** to archive
- Propose new tests based on recent JIRA epics/stories

4. Test Execution Analysis

- Parse Jenkins test results and summarize **failures by module**
- Suggest RCA hints using historical defect data
- Generate **smart dashboards** for QA sprint reviews

5. Defect Triage & RCA Suggestions

- Classify bugs by severity/priority
- Recommend likely impacted features (based on past JIRA logs)
- Auto-generate RCA templates using historical fixes

6. KT & Onboarding Assistance

- Serve as a QA knowledge bot: “How to set up PRIME framework?”
- Link internal SOPs, shared drive content, and qTest assets

7. Cross-Tool Integrations

- SASVA connectors to:
 - qTest API for test case lookup, updates
 - GitLab Merge Requests for test coverage validation
 - Jenkins REST API for test results and failure logs
 - BrowserStack for screenshot analysis

Section 3: Phased Implementation Plan

Phase	Timeframe	Key Focus
Phase 1	April 2025	Enable SASVA to co-author test cases and automation scripts
Phase 2	May–June 2025	Use SASVA for test results summarization & defect RCA
Phase 3	Q3 2025	Autonomous test case validation, flaky test alerting, and integration into QA dashboards

Section 4: Real Examples of SASVA in Action

Scenario	What SASVA Would Do
A new login page is released	Suggests 8+ test cases in Given-When-Then format with validations
Jenkins reports 6 failed tests	Provides RCA hints: stale locators, DB validation fail, or broken test data
Sprint review in 2 hrs	SASVA generates a summary of passed/failed/smoke/regression tests by module
Test engineer is onboarded	Can ask: “Where’s the SQL for Sub Bene withdrawal validation?” SASVA returns link instantly

Section 5: GenAI Comparison Table

Feature / Tool	SASVA (Persistent)	GitHub Copilot	ChatGPT (OpenAI)	Other Tools (Claude, Gemini)
Contextual QA Intelligence	Deep QA domain-specific training	General coding only	General NLP	General NLP
STLC Awareness	Test design to triage	None	Manual prompt needed	Manual prompt needed
Integration with Persistent Ecosystem	Seamless (SaaS/SharePoint/JIRA /qTest)	Needs setup	No native access	
Cucumber/TestNG/BDD Coding	Pre-trained examples	Code only	Requires prompting	
Internal SOP & Shared Drive Access	YES			
Conversational Task Automation	Task-based prompts with voice			
QA Report Generator	Yes		With advanced prompting	
Ideal Use Case	QA Automation/Analysis	Code Suggestions	General Research	Q&A / Research

Verdict:

SASVA wins as the best choice for QA Engineering within Persistent’s ecosystem due to its:

- Secure access to internal tools
- Built-in knowledge of automation stack
- Project-level contextual memory
- Goal of making every QA a GenAI-augmented contributor

Summary: Why SASVA Should Be Your Virtual QA Engineer

SASVA is not just an assistant — it is your **AI QA Co-Engineer**:

- Trained in Persistent terminology, repositories, and test stacks
- Reduces QA backlog, triages bugs faster, and keeps test suites lean and healthy
- Empowers QA with onboarding help, KT, coverage reports, and test generation